NYC Taxi Fleet Optimizer: Predicting Hourly Urban Demand with Weather Integration

Maxim Lavelin
Bar-Ilan University
maxim.lavelin@live.biu.ac.il

Abstract

Accurate short-term demand forecasting is essential for improving the efficiency of urban taxi fleets. In New York City, strong temporal variability in taxi demand leads to excessive idle cruising during low-demand periods and service shortages during peak hours. This project addresses the problem of predicting hourly, citywide taxi demand using supervised regression models trained on historical ride data and weather information. NYC TLC yellow taxi trip records aggregated at an hourly level are combined with corresponding weather observations from JFK Airport. An interpretable linear regression model is used as a baseline and compared against a non-linear XGBoost model. Evaluation follows a strict chronological split to reflect real-world deployment. Results show that while linear regression captures overall seasonality, it systematically underestimates peak demand. XGBoost significantly improves performance, achieving an approximate 25% reduction in Mean Absolute Error by leveraging temporal structure and historical demand memory. The findings demonstrate the operational value of non-linear models for urban fleet optimization.

1 Introduction

Urban taxi systems operate under highly dynamic demand conditions driven by time-of-day patterns, weekly cycles, and external factors such as weather. In New York City, inaccurate demand anticipation results in substantial inefficiencies: drivers waste fuel and time cruising without passengers during low-demand periods, while passengers experience long wait times and reduced service quality during demand peaks.

This project formulates hourly taxi demand forecasting as a supervised regression problem, where the goal is to predict the total number of taxi pickups across New York City for a given hour. Accurate forecasts can support proactive fleet positioning, reduce idle mileage, improve driver revenue, and mitigate congestion and emissions.

The main contributions of this work are threefold. First, a clean, operationally relevant dataset is constructed by merging hourly taxi demand with weather observations. Second, an interpretable linear regression baseline is compared against a more expressive XGBoost model under a realistic evaluation protocol. Third, detailed visualization and error analysis are used to identify where and why non-linear models provide value, particularly during peak-demand periods.

2 Data & Exploratory Analysis

The dataset used in this study combines New York City taxi demand data with corresponding meteorological observations to construct an hourly, citywide demand forecasting problem. Taxi demand data were obtained from the NYC Taxi and Limousine Commission (TLC) Yellow Taxi Trip Records, publicly available at: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page. Trip-level records were aggregated to hourly pickup counts across the entire city for the year 2024. The resulting target variable, *Ride_Count*, represents the total number of taxi pickups per hour in New York City.

The final modeling dataset contains 8,616 hourly observations. Summary statistics for *Ride_Count* reveal substantial variability in demand: the mean hourly demand is 4,707.52 rides with a standard deviation of 2,711.53. Demand ranges from a minimum of 0 rides during rare off-peak hours to a maximum of 12,757 rides during extreme peak periods. This wide range highlights the inherently volatile nature of urban taxi demand and motivates the need for models capable of handling sharp transitions and high variance.

Weather data were sourced from the National

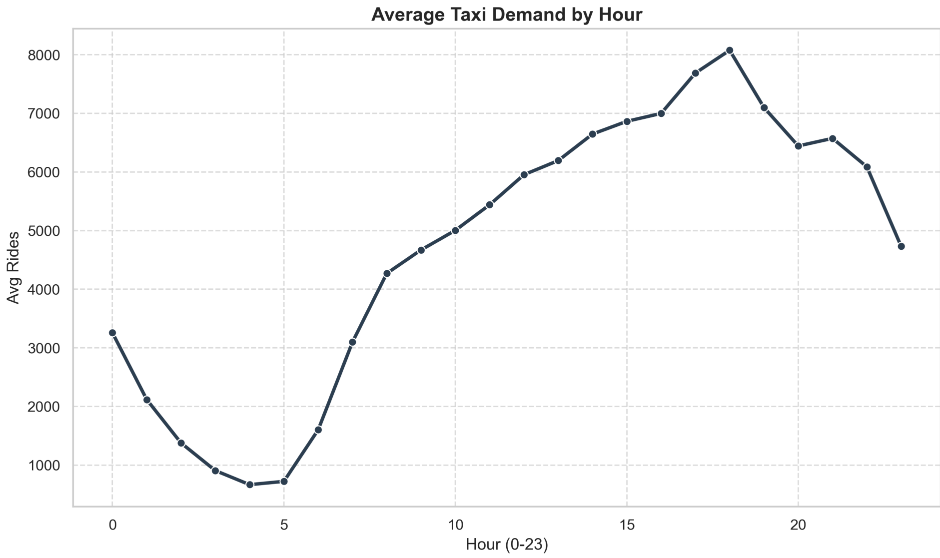

Figure 1: Average Daily Ride Volume

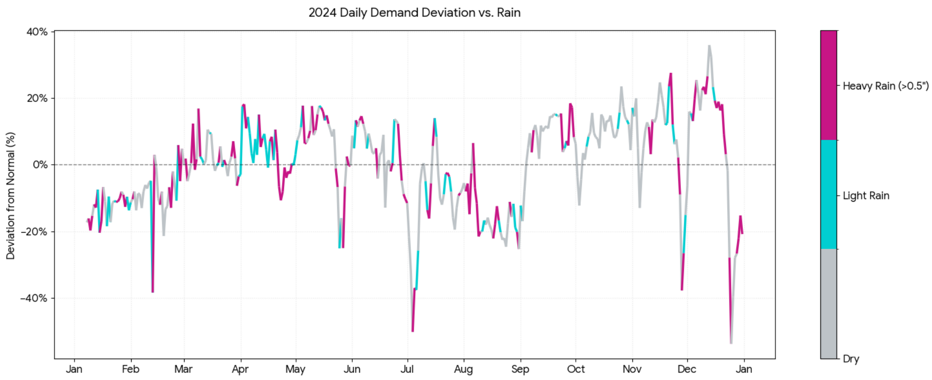

Figure 2: 2024 Weather Impact on Demand

Centers for Environmental Information (NCEI) Local Climatological Data archive for JFK Airport: https://www.ncei.noaa.gov. The raw weather dataset consists of over 13,000 observations with approximately 125 meteorological features spanning hourly, daily, and monthly measurements. As part of preprocessing, the data were aggregated to an hourly resolution to align with the taxi demand series and then reduced to a limited subset of weather variables relevant for demand prediction.

Exploratory analysis guided feature selection, prioritizing operational relevance and interpretability. Temporal structure was explicitly modeled using hour-of-day, day-of-week, and weekend indicators, reflecting strong recurring demand cycles. Weather variables were intentionally simplified: air temperature was retained as a continuous feature, while precipitation was transformed into a binary rain indicator classifying each hour as wet or dry.

Summary statistics for the selected weather features show that hourly temperature ranges from $-10.6\,°\mathrm{C}$ to $34.4\,°\mathrm{C}$, with a mean of $14.05\,°\mathrm{C}$ and a standard deviation of $8.86\,°\mathrm{C}$. Precipitation exhibits a highly skewed distribution, with a mean of only 0.13 mm, a standard deviation of 0.74 mm, and a maximum hourly value of 18.5 mm. These distributions support the decision to encode precipitation as a binary indicator, as demand response was found to depend more on the presence of rain than on its precise intensity.

As shown in Figure 1 (Page 2), the daily ride volume reveals pronounced morning and evening peaks and a stable diurnal pattern. Figure 2 (Page 2) illustrates the relationship between weather and demand during 2024, indicating that rain alone does not consistently increase or decrease taxi demand. These findings reinforce the need to model weather effects jointly with temporal and historical demand features rather than in isolation.

3 Baseline Model — Linear Regression

The baseline model is formulated as a multiple linear regression, in which hourly taxi demand is modeled as a linear combination of engineered temporal, weather, and historical features. Formally, the model assumes the relationship:

$$\hat{y}_t = \beta_0 + \sum_{i=1}^{p} \beta_i \, x_{i,t} \tag{1}$$

where $\hat{y}_t$ denotes the predicted number of taxi pickups at hour t, $x_{i,t}$ represents the input features, and β_i are coefficients estimated via ordinary least squares by minimizing the sum of squared residuals. This formulation assumes linearity, additive feature effects, and constant variance of errors.

Figure 3 (Linear Regression Example) illustrates this modeling assumption. The fitted regression line captures the average relationship between the independent variable and demand, while individual observations deviate around the line due to unmodeled variability. This visualization highlights the primary strength of linear regression: transparency and interpretability.

In practice, the linear model captures broad seasonal and diurnal patterns through temporal indicators and lagged demand features. However, its strictly additive structure prevents it from modeling non-linear interactions and regime-dependent behavior. As a result, the model exhibits systematic errors during peak-demand periods, where demand increases rapidly and real-world dynamics violate linear assumptions. These limitations motivate the use of a more expressive non-linear model.

4 Improved Model — XGBoost

To overcome the structural limitations of linear regression, an XGBoost regressor is employed as the improved model. XGBoost is a gradient-boosted ensemble of decision trees that models the target variable as an additive expansion of weak learners:

$$\hat{y}_t^{(k)} = \sum_{j=1}^{k} f_j(x_t), \quad f_j \in \mathcal{F} \tag{2}$$

where each f_j is a regression tree and $\mathcal{F}$ denotes the space of tree functions. Trees are added sequentially to minimize a regularized objective consisting of a loss term (squared error) and a complexity penalty that constrains tree depth and leaf weights.

Figure 4 (XGBoost Scheme) illustrates the boosting process. An initial prediction is produced, residual errors are computed, and successive decision trees are trained to predict these residuals. Each iteration updates the model by adding a scaled contribution from the new tree, gradually refining predictions. The learning rate controls update magnitude, ensuring stable convergence and improved generalization.

This iterative error-correction mechanism enables XGBoost to learn non-linear relationships and conditional interactions between features, such as demand responding differently to weather depending on hour-of-day or recent demand history. Regularization reduces overfitting, while boosting concentrates modeling capacity on difficult-to-predict regimes, particularly peak-demand periods.

5 Evaluation and Results

Model evaluation followed a strictly chronological data split to reflect real-world forecasting conditions and prevent information leakage. Data from January through September 2024 were used for training, October served as a validation period for hyperparameter tuning, and November–December 2024 were reserved for final out-of-sample testing. This setup mirrors an operational deployment scenario in which future demand must be predicted solely from past observations.

Mean Absolute Error (MAE) was selected as the primary evaluation metric due to its direct interpretability in terms of rides per hour. Root Mean Squared Error (RMSE) was also reported to emphasize large prediction errors, which are particularly costly during peak-demand periods and disproportionately affect fleet allocation decisions.

Final test results on the November–December 2024 period demonstrate a clear performance gap between models. The linear regression baseline achieved an MAE of 728.43 rides per hour and an RMSE of 993.47. In contrast, the tuned XGBoost model achieved an MAE of 546.19 and an RMSE of 810.55. This corresponds to an absolute MAE reduction of approximately 182 rides per hour, representing an improvement of roughly 25%.

Figure 5 (Actual Demand vs. Model Projections Comparison) visually confirms these quantitative results. The linear regression model cap-

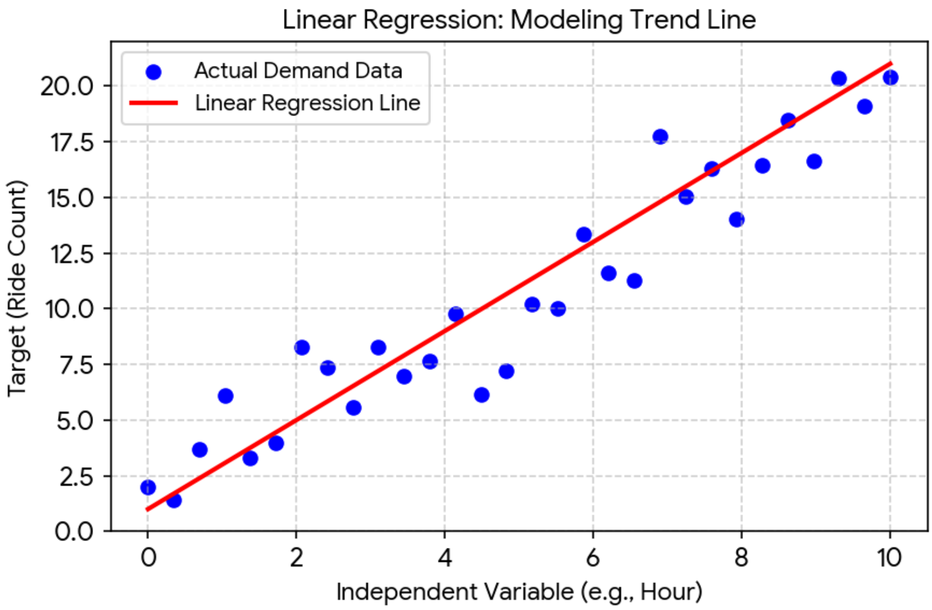

Figure 3: Linear Regression Example

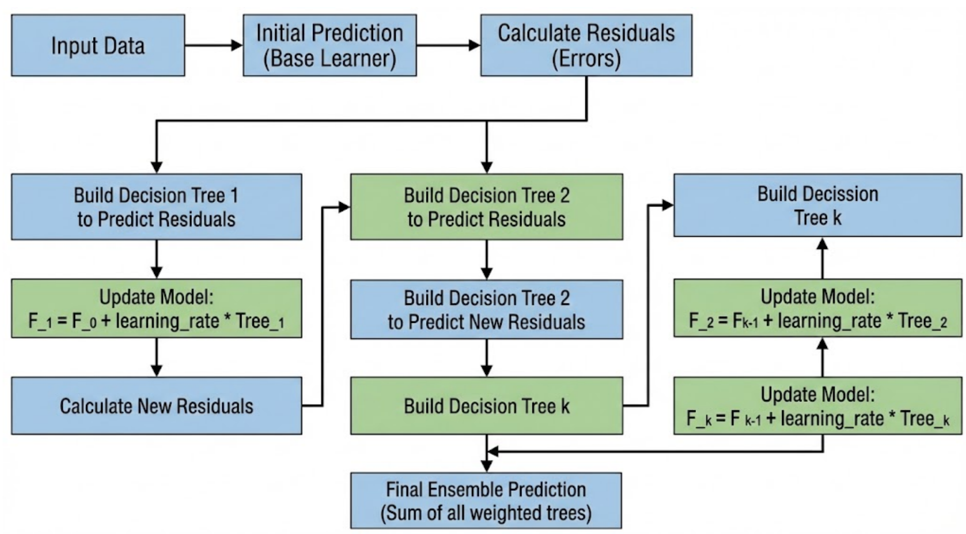

Figure 4: XGBoost Scheme

tures overall seasonality but consistently underestimates sharp demand peaks and rapid transitions. XGBoost, by contrast, closely tracks both the magnitude and timing of observed demand fluctuations. Figure 6 (XGBoost Feature Importance) further explains this improvement, showing that lagged demand features and hour-of-day indicators dominate predictive performance, with weather variables contributing secondary contextual information.

6 Visualization & Error Analysis

Time-series visualization provides insight into how prediction errors vary across time rather than being uniformly distributed. As shown in Figure 5, the linear regression model exhibits its largest deviations during peak hours, where demand rises sharply and linear assumptions break down. These errors are particularly costly from an operational perspective, as peak-hour mispredictions lead to increased passenger wait times, inefficient fleet allocation, and lost revenue.

XGBoost significantly reduces error during these high-impact periods, closely tracking demand peaks and rapid transitions. Feature importance analysis in Figure 6 highlights the dominant role of lagged demand variables, emphasizing the impor-

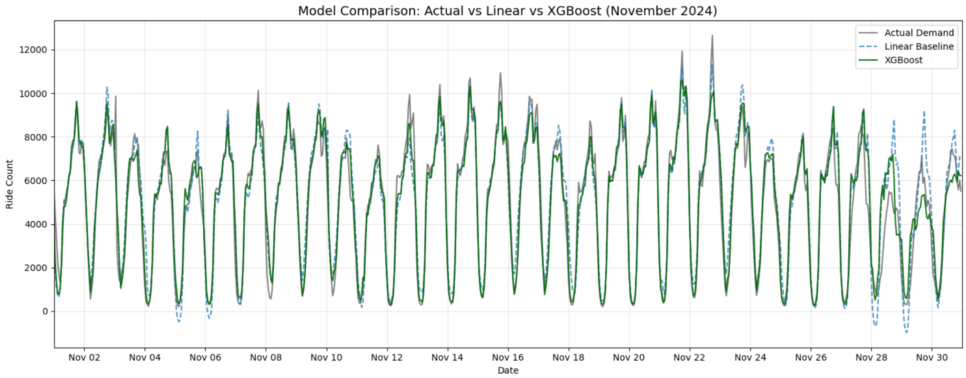

Figure 5: Actual Demand vs. Model Projections

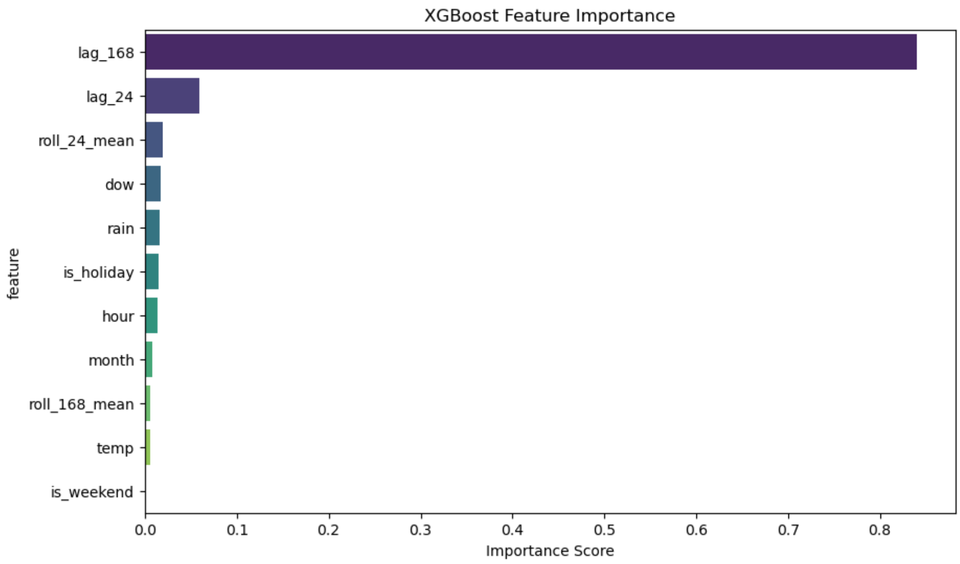

Figure 6: XGBoost Feature Importance

tance of short-term memory in forecasting. This suggests that improving representations of historical demand—through refined lag structures or adaptive rolling statistics—is likely to yield greater accuracy gains than adding raw external variables alone.

Visualization therefore functions not only as diagnostic analysis but also as guidance for future model development, demonstrating that the benefits of XGBoost are concentrated precisely where accurate forecasting is most valuable.

7 Conclusions Future Work

This project demonstrates that hourly taxi demand in New York City is governed primarily by strong temporal regularity and short-term historical dependence. Exploratory figures show stable daily and weekly cycles, while weather effects—particularly rain—exhibit inconsistent and non-monotonic influence. These characteristics explain why the linear baseline fails most visibly during peak hours (Figure 3), where demand changes rapidly and nonlinear relationships dominate.

The XGBoost model substantially improves forecasting accuracy, achieving roughly a 25% reduction in MAE. Feature importance analysis confirms that lagged demand and hour-of-day features are the dominant predictors, with weather variables playing a secondary role as contextual modifiers. This validates the modeling approach and highlights the value of nonlinear methods for urban demand forecasting.

Several limitations remain. Weather data from a single airport station provides only a coarse proxy for citywide conditions, city-level aggregation obscures spatial heterogeneity across neighborhoods, and rare or non-recurring demand spikes remain difficult to predict. Future work includes integrating hyper-local weather data from multiple stations, developing zone-level demand models for high-density areas such as Manhattan, and incorporating event data (e.g., concerts and sports games) to better capture non-seasonal demand shocks and further reduce prediction error.

8 Team Member Roles

This project was completed individually. All tasks, including data collection, preprocessing, modeling, evaluation, visualization, and report writing, were performed by a single contributor.